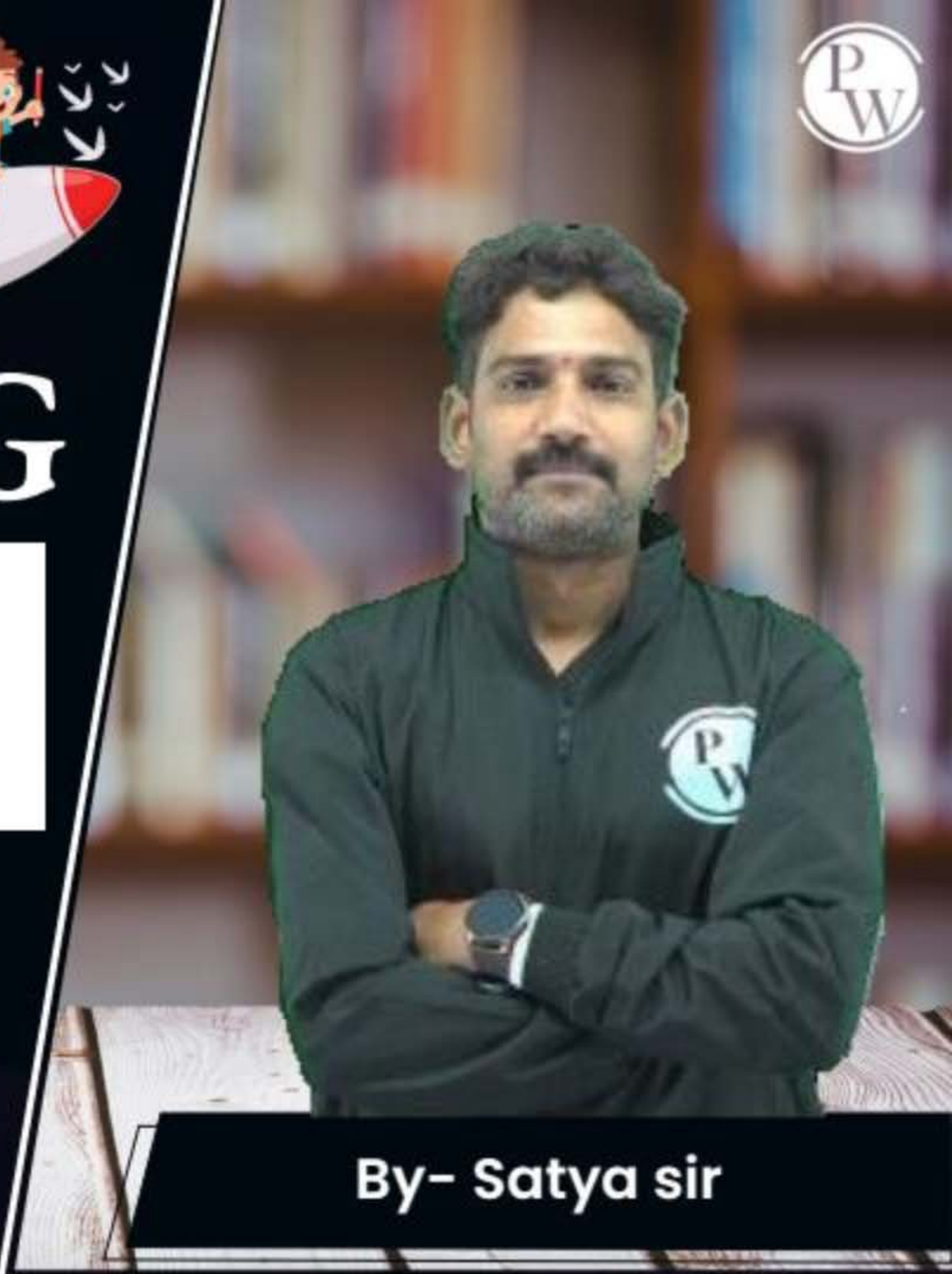
CS & IT ENGINEERING

Data Structure & Programming

Linked Lists

Lecture No.- 01



By- Satya sir

Recap of Previous Lecture

- Arrays
  - 1-D array
  - 2-D array
  - 3-D array

  } Initialization / Assignment

- Address of an Element with random index
- Searching an Element in an array
  - Linear Search
  - Binary Search

Topics to be Covered

- Array Insertion
- Array Deletion
- Linked Lists
  - Characteristic
  - Types
    - Construction
    - Insertion

} SLL

Topic : 1-D Arrays – ISRO 2009

A one dimensional array A has indices 1....75. Each element is a string and takes up three memory words. The array is stored at location 1120 decimal. The starting address of A[49] is ____

B = 1120 Starting index, x = 1, Size, n = 3

$$A[i] = B + (i - x) * n$$

$$A[49] = 1120 + (49 - 1) * 3$$

$$= 1120 + 48 * 3$$

$$= 1120 + 144$$

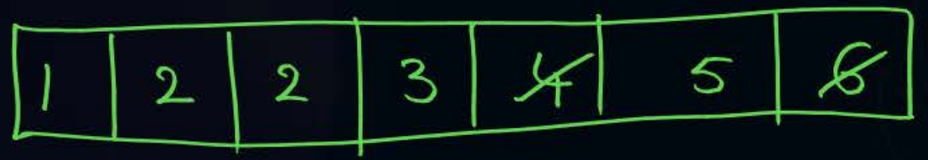
$$= 1264$$

Topic : 1-D Arrays – GATE 2024

Let A be an array containing integer values. The distance of A is defined as the minimum number of elements in A that must be replaced with another integer so that the resulting array is sorted in a non-decreasing order. The distance of the array [2, 5, 3, 1, 4, 2, 6] is 3

Given Array

Result Array



Length of given array = 7

Length of LIS = 4

⇒ (7 − 4) = 3 Elements need to sorted.

Longest Increasing Sub Sequence

[2, 3, 4, 6]

2 5 3 1 4 2 6

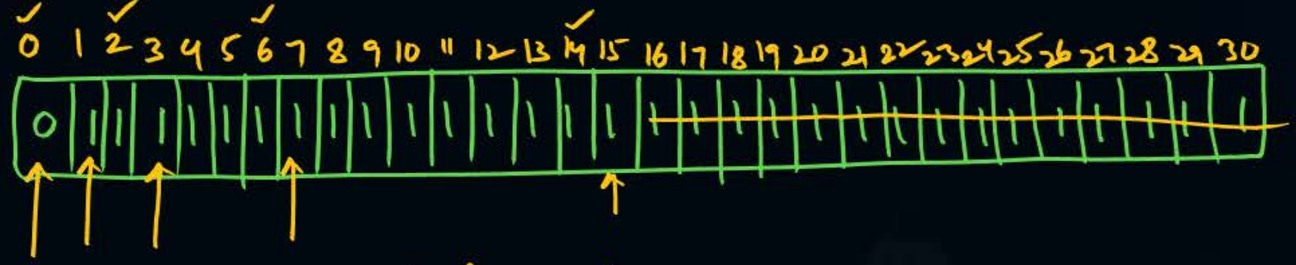
IS: [2 5 6] length = 3

IS: [5 6] length = 2

	length
IS: [1 4 6]	3
IS: [2, 3, 4, 6]	4
IS: [1, 2, 6]	3
IS: [3, 4, 6]	3
IS: [4, 6]	2
⋮	

Topic : 1-D Arrays – GATE 2017

Let A be an array of 31 numbers consisting of sequence of 0's followed by a sequence of 1's. The problem is to find the smallest index i that A[i] is 1 by probing the minimum numbers of locations in A. The worst-case number of probes performed by an optimal algorithm is _____.



Possibilities

000000...0000...11111

00000000000000001111111111111111

$mid = \frac{low + high}{2} = 15, 7, 3, 1$

Worst-case

01111111111111111....1

00000.....01

(Worst case) Linear Search : 31 Comp

Binary search (Worst case) : $\lceil \log_2 31 \rceil = 5$ Comp

(Best Case) Linear Search : 2 comparisions 1 is found, Smallest index also found.

Binary Search (Worst Case) : $\lceil \log_2 31 \rceil$ comparisions $= 5$

Topic : 1-D Arrays – ISRO 2018

An array A consists of n integers in locations A[0], A[1] ....A[n-1]. It is required to shift the elements of the array cyclically to the left by k places, where 1 <= k <= (n-1). An incomplete algorithm for doing this in linear time, without using another array is given below. Complete the algorithm by filling in the blanks. Assume all the variables are suitably declared.

```
min = n; i = 0;
while (_____) {
    temp = A[i]; j = i;
    while (_____) {
    A[j] = _____
    j= (j + k) mod n ;
    If ( j< min ) then
       min = j;
}
A[(n + i — k) mod n] = _____
i = _____
```

Let

0	1	2	3	4
1	3	8	7	9

5 7 9 1 3

i=0 temp=1 j=0

0!=3 A[0]=A[2]

j=2 2<5 True

min=2

2!=3 True

A[2]=A[4]

j=4

4!=3 True

A[4]=A[1]

j=1

1!=3 True

A[1]=A[3]

j=3

3!=3 False

A[3]=1

i=2

2<2 False

K = 2 Result

5	7	9	1	3

A. i > min; j!= (n+i)mod n; A[j + k]; temp; i + 1 ;

B. i < min; j!= (n+i)mod n; A[j + k]; temp; i + 1;

C. i > min; j!= (n+i+k)mod n; A[(j + k)]; temp; i + 1;

D. i < min; j!= (n+i-k)mod n; A[(j + k)mod n]; temp; i + 1;

Topic : Arrays Insertion, Deletion

Array - Insertion / Add an Element

Let Array[10] = {1,3,5,7} count = 4

Worst-case : Insert at first Position

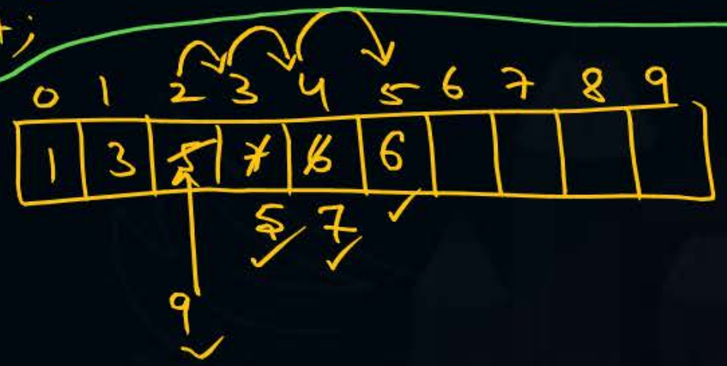
Value = 8 at First Position = index 0

```
i = count // 6
while( i > 0)
{ Array[i] = Array[i-1]
  i = i-1
} Array[i] = value; Count++
```

n times ⇒ Time Complexity O(n)

Best case: Insert at End of list Array[Count] = value; Count++; Time Complexity = = O(1)

Let value = 6



Average case : Insert at middle of list Array

Position = 2, value = 9

i = Count // i = 5

```
while( i > Position)
{ Array[i] = Array[i-1]
  i = i-1
}
Array[Pos] = value;   Count++;
```

n/2 times ⇒ Time Complexity : $O(\frac{n}{2}) \Rightarrow O(\frac{1}{2} * n) \Rightarrow O(n)$

Topic : Linked Lists

Deletion from Array / Remove an Element

Best case : Last Element

Let A = [1, 3, 5, 7, 9] count = 5

Printf(" Removed Element is %d", A[--count]); // Time Complexity = O(1)

Average Case : Deletion of middle element : Time Complexity : $O(\frac{n}{2}) \Rightarrow O(n)$

⇒ Shift Elements right of middle to left

A[i] = A[i+1]

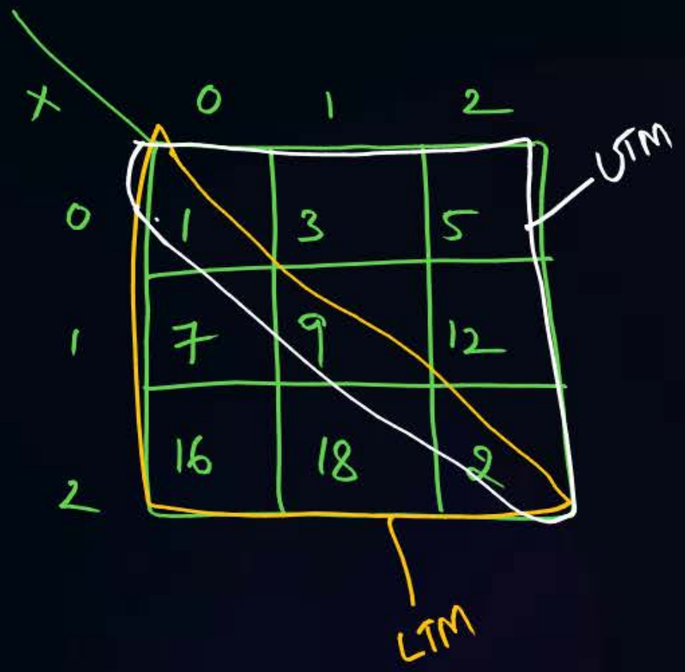
Worst-Case : Delete first element : Shift all elements left by 1 Position ⇒ Time Complexity : O(n)

Topic : SLL – Creation, Insertion

2-D Arrays – LTM, UTM

n = No. of rows / cols = 3 [Square Matrix]

Let x[3][3] = {1, 3, 5, 7, 9, 12, 16, 18, 2}



LTM

1	0	0
7	9	0
16	18	2

UTM

1	3	5
0	9	12
0	0	2

No. of Elements = 6

= 6

$$\Rightarrow \boxed{\frac{n(n+1)}{2}}$$

n = No. of rows (or) cols in Square Matrix

Topic : SLL – Creation, Insertion

Linked Lists

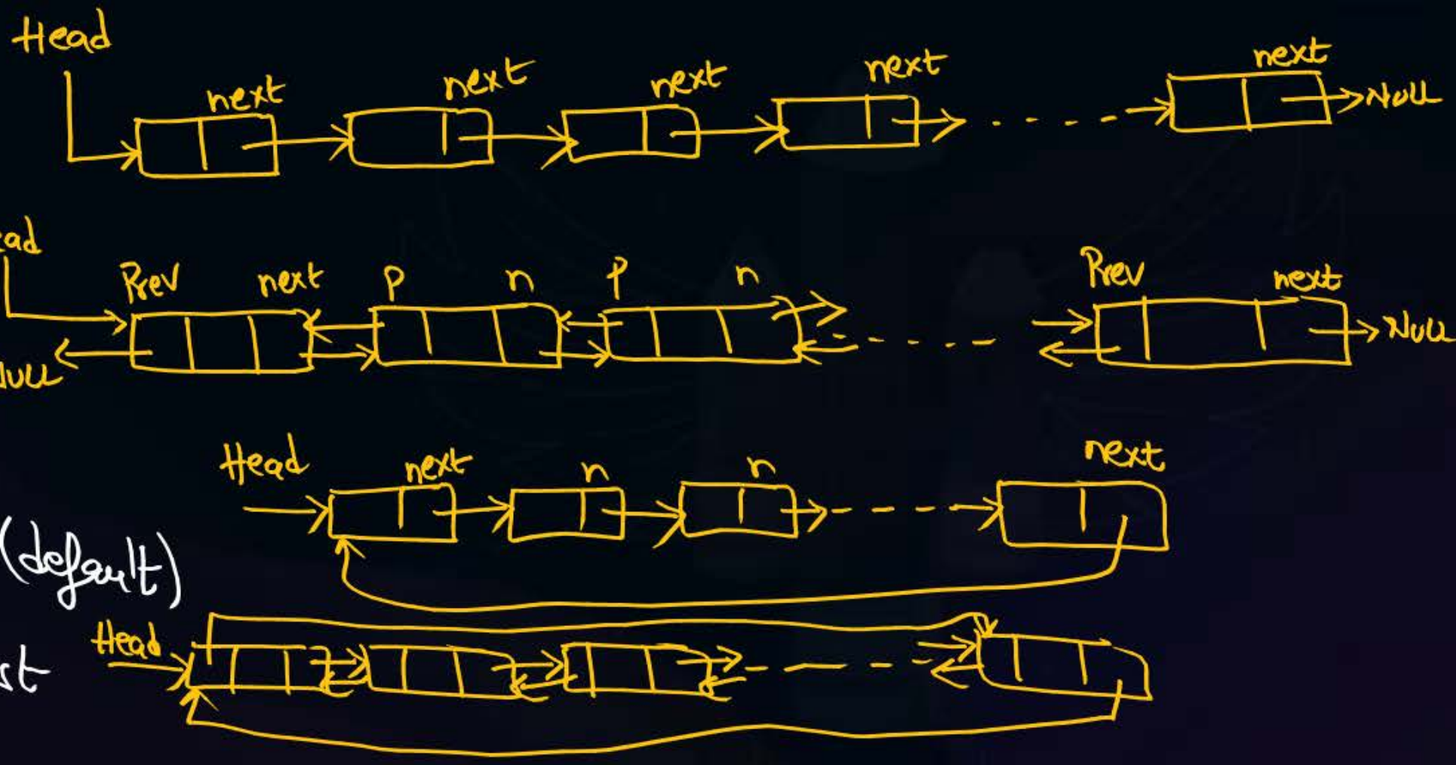
- List (or) group of Elements that are linked together.

- 3 Types of Linked Lists :

  1) Singly Linked List (SLL)

  2) Doubly Linked List (DLL)

  3) Circular Linked List (CLL)
     - Circular Singly Linked List (default)
     - Circular doubly Linked List



Topic : SLL – Creation, Insertion

- Create Node
- Insert Node(s)
- Construct Linked List
- Delete an Element
- Search for an Element
- Access (or) Traverse LL

To be discussed...

2 mins Summary

- H/W Questions Solving
- Array Insertion, Deletion
- Array - LTM, UTM
- Linked Lists Introduction

To be contd . . .

THANK - YOU